

		$\frac{1}{2} \frac{2R}{\left(\frac{1}{2} R \right)} = \frac{1}{2} \left(\frac{R}{2R} \right)$
13	A	The frictional force is the restoring force. $\sum F = ma = m\omega^2 x$ $5.0 = 0.20 \times (2\pi \times 1.5)^2 x_0$ $x_0 = 0.28 \text{ m}$
14	B	For a closed pipe the resonant lengths are odd multiples of the shortest length